

DCS — Pinout & Terminal Connections

Pi 4 GPIO · level shifter · ADS1115 · 250 Ω shunt · Rosemount loop

RASPBERRY PI 4 (BCM) via GPIO screw TB

Pin 1 (3V3) → LLC LV
Pin 3 (GPIO2/SDA) → LLC A1 (SDA)
Pin 5 (GPIO3/SCL) → LLC A2 (SCL)
Pin 6 (GND) → LLC GND · ADS GND · shunt low
Pin 11 (GPIO17) → Relay IN (optional)
LLC HV / B1 / B2 → ADS VDD(5V) / SDA / SCL
I2C bus 1; ADS ADDR→GND ⇒ 0x48

ADS1115 + 250 Ω SHUNT

VDD ← 5 V (not Pi 3V3)
GND ← star at shunt low
SCL/SDA ← LLC HV side
AIN0 ← shunt high
Rs = 250 Ω 0.1%
4 mA→1.00 V 20 mA→5.00 V
PGA ±6.144 V recommended at 5 V FSR

4–20 mA LOOP

+24 VDC (Mean Well DIN) → Rosemount +
Rosemount – → shunt high → ADS1115 AIN0
shunt low → 24 V– AND ADC/Pi GND (single star at brass ground bar)
Verify with Fluke 789 mA OUTPUT: 8 mA ⇒ ~2.00 V / 25% span on terminal stream

ASSEMBLY

Wood-board / junction-box: Pi 4 + green perfboard HAT with blue 4 Bi-Directional Level Shifters + DIN Mean Well + brass ground bar.
Hero photo: docs/images/photo_assembled_stack.png · Photoreal composite: docs/images/dcs_bom_junction_box_illustration.png
